Research and Evaluation of Multilingual Speech and Language Models

INTERNSHIP REPORT

Submitted By

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RESILIENCE BUSINESS GRIDS LLP, COIMBATORE

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VATTAMALAIPALAYAM, N.G.G.O. COLONY POST,

COIMBATORE – 641 022.

January 2026

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BONAFIDE CERTIFICATE

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Place: Coimbatore

Date:

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We extend our sincere gratitude to all the teaching and non-teaching staff members of our department who helped us during our internship.

1. INTRODUCTION

Throughout my university years, I gained valuable theoretical knowledge from lectures, library books, journals, and other secondary resources as part of my graduation studies. However, these were mostly conceptual rather than hands-on. Internships bridge that gap by applying theory to real-world practice, which is why they hold such importance.

Internships offer the ideal opportunity to understand workplace dynamics, company culture, and how academic concepts translate into practical projects. I feel incredibly lucky to have completed my internship at the renowned **Resilience Business Grid LLP**.

As an intern there, I collaborated closely with the team on their projects and contributed meaningfully. One highlight was the chance to work alongside brilliant minds, which sharpened both my technical expertise and interpersonal abilities. Overall, my time at Resilience Business Grid LLP was an enriching experience.

1. OVERVIEW OF THE ORGANIZATION

Resilience Business Grid provides the Additive Artificial Intelligence Service Stack (AISS), a dynamic continuous learning platform that integrates plug-and-play microservices customized to your exact needs for sustainable business growth through our AI solutions and domain-expert consultants. It features deep learning models for tailored NLP solutions that extract insights from extensive text data, alongside vision analytics to build AI systems capable of perceiving, analyzing, learning, and reasoning from images and videos. The stack simplifies human-machine interfaces with powerful deep learning-based speech solutions compatible with any infrastructure, while offering data-driven big data services to enable reliable IoT via transparent descriptive, predictive, and prescriptive analytics models.

Office	RBG.AI, SREC SPARK Incubation Foundation, Sri Ramakrishna Engineering College, Coimbatore – 641022, India.
Type of the organization	Artificial Intelligence Research organization
Years of Experience	4 Years (incorporated in 2021)

Nature of business	Resilience Business Grid LLP Brand Consulting Services encompass both Technology & Artificial Intelligence Research Solutions creating a holistic package to take your Brand where it deserves to be.
Business motto	"We provide the services to structure the unstructured data with machine learning and deep learning technology"
URL (website)	https://rbg.ai/

No day is a regular or boring day at Resilience Business Grid. Every day is vibrant, dynamic and different. As employee engagement program, RBG.AI organizes timely outings, excursions, and ceremonies.

2. ABOUT MY PROJECT

During my internship at Resilience Business Grid (RBG.AI), I concentrated on researching and evaluating speech and language processing technologies, with an emphasis on machine translation for Indian languages. This entailed a thorough assessment of multilingual language models to gauge their capability in managing India's diverse linguistic landscape.

The core project goal was to investigate and benchmark machine translation models from the Hugging Face repository, comparing various English-to-Indian language and bidirectional models based on their precision and efficiency. I reviewed numerous research papers and technical resources, prioritizing large-scale transformer architectures featuring extensive parameter counts.

Translation performance was measured using datasets across multiple Indian languages, with model outputs evaluated via the BLEU score, the established benchmark for translation accuracy.

In the internship's later phase, I gathered and structured speech datasets for diverse Indian languages to aid upcoming research, while also compiling collections of authentic and synthetic voice samples from prominent Indian figures to advance studies in voice verification and deepfake detection.

SPEECH AND LANGUAGE PROCESSING:

Speech and Language Processing represents a key AI domain dedicated to empowering computers to comprehend, process, and produce human language. It integrates concepts from computer science, linguistics, and statistical methods to handle both text and spoken data. As digital interactions and multilingual communities expand rapidly, this field proves vital for overcoming language obstacles and enhancing information access.

Recent progress in deep learning and transformer architectures has dramatically boosted the effectiveness of these systems, enabling them to manage intricate linguistic patterns, contextual nuances, and diverse multilingual datasets. For a linguistically varied nation like India, robust speech and language processing tools are indispensable for facilitating seamless communication across numerous languages and dialects.

MACHINE TRANSLATION MODELS:

Machine Translation serves as a fundamental application within speech and language processing, aimed at automatically converting text between languages while maintaining its intended meaning. Contemporary systems rely on transformer-based encoder-decoder frameworks trained on vast multilingual corpora. These prove especially vital for Indian languages, most of which qualify as low-resource owing to scarce high-quality training data.

In the project, I examined and evaluated machine translation models hosted on the Hugging Face platform, with particular focus on those supporting English-to-Indian language and reverse-direction translations—like expansive transformer models boasting substantial parameter counts. Such models exhibited enhanced multilingual adaptability and superior translation accuracy across varied Indian languages. Relevant research papers and technical documents were reviewed to grasp their architectures, training methodologies, language coverage, and performance metrics.

MODEL EVALUATION:

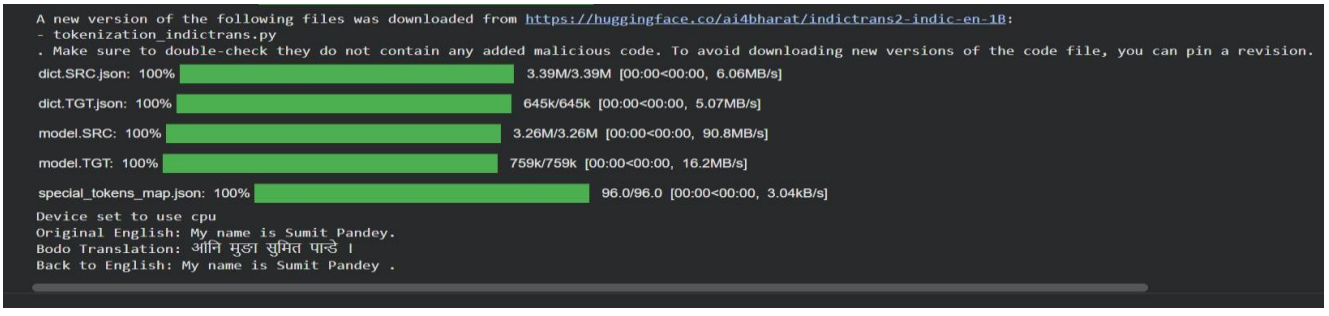
Assessing machine translation models' performance is an essential process for determining their accuracy and dependability. In this project, the BLEU (Bilingual Evaluation Understudy) score—a standard metric for translation quality—was applied to evaluate the models. This score measures how closely machine-produced translations align with reference versions by examining similarities at the word and phrase levels.

3. COMPONENTS

- COLAB-BASED MODEL EVALUATION
- INDIAN LANGUAGE TRANSLATION BENCHMARKING SYSTEM
- MULTILINGUAL VOICE DATASET PIPELINE
- DEEPFAKE DETECTION DATASET ARCHITECTURE
- TRANSFORMER MODEL ANALYSIS MODULE

COLAB-BASED MODEL EVALUATION:

This section describes the setup employed for running experiments and assessments on machine translation models. Google Colab served as the main platform, chosen for its cloud-hosted environment, access to GPU/TPU resources, and easy compatibility with Python machine learning libraries. Colab facilitated smooth loading, testing, and benchmarking of large transformer models without needing powerful local hardware. It also enabled quick iterations and consistent result reproduction during evaluations.



INDIAN LANGUAGE TRANSLATION BENCHMARKING SYSTEM:

The Indian Language Translation Benchmarking System focuses on evaluating machine translation models for English-to-Indian and Indian-to-English language pairs. This component involved comparing multiple multilingual translation models available on the Hugging Face platform. The benchmarking process assessed model performance across different Indian languages, considering linguistic diversity, script variations, and resource availability. This system helped identify models that performed effectively for both high-resource and low-resource Indian languages.

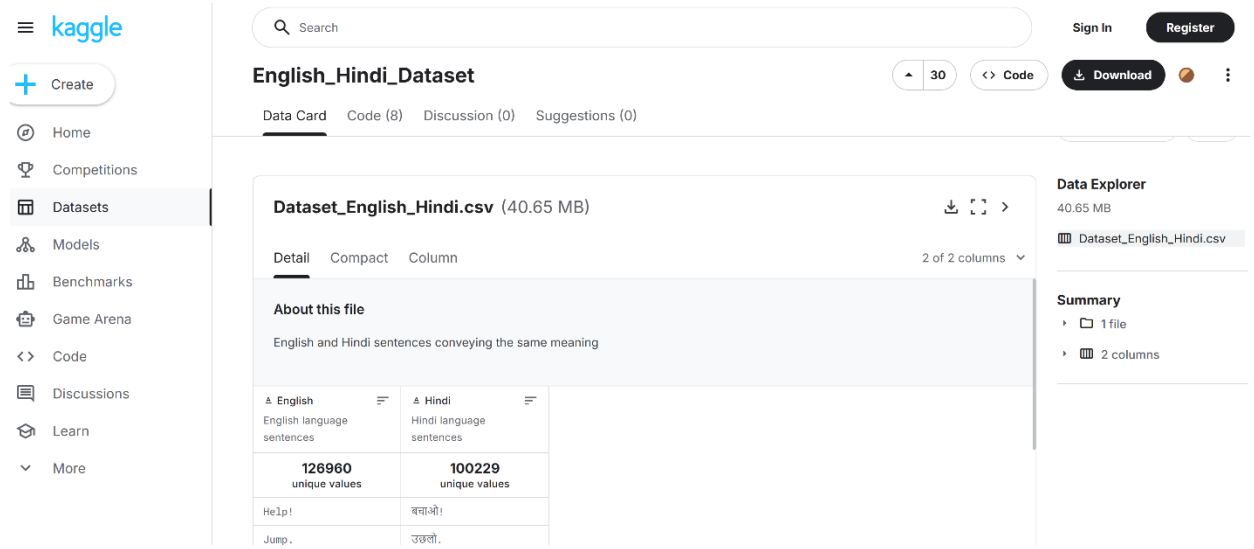
```
# Output
print("Original English:", text)
print("Manipuri Translation:", intermediate_result)
print("Back to English:", reverse_result)
```

```
Device set to use cpu
Device set to use cpu
Original English: My name is Sumit Pandey.
Manipuri Translation: ꯑꯩꯒꯤ ꯃꯤꯡ ꯁꯨꯃꯤꯠ ꯄꯥꯟꯗꯦ ꯁꯤꯔꯣꯛꯄꯤ ꯔ
Back to English: My name is Sumit Pandey .
```

MULTILINGUAL VOICE DATASET PIPELINE:

This section covers the gathering, structuring, and preprocessing of speech datasets across various Indian languages. A systematic workflow was implemented to source voice recordings from diverse platforms and format them for upcoming research applications.

Data was sourced from repositories like Mozilla Common Voice, OpenSLR, Kaggle, and open-source GitHub collections. Common Voice provided a primary multilingual resource with broad coverage of global and Indian languages. OpenSLR offered premium, vetted speech corpora favored in academic studies, Kaggle supplied organized sets for targeted languages, and GitHub hosted specialized datasets from academic and industry contributors.

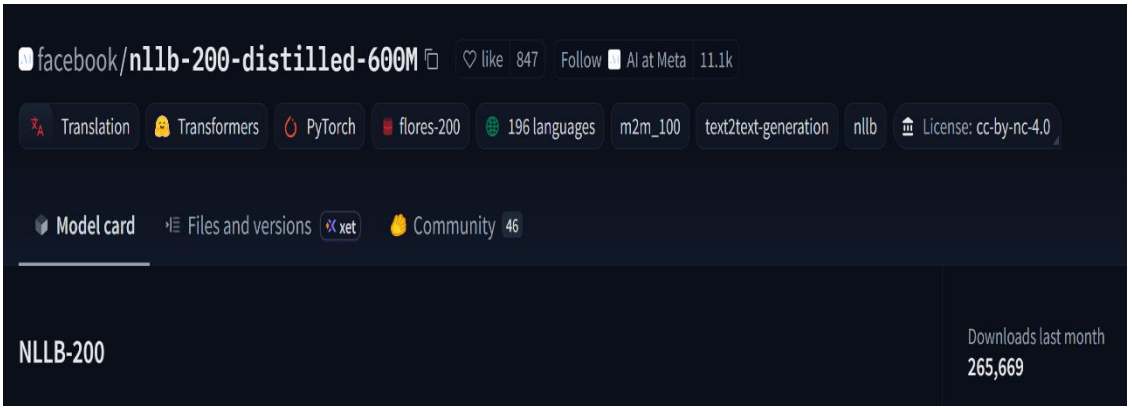


DEEPFAKE DETECTION DATASET ARCHITECTURE:

The Deepfake Detection Dataset Architecture component focuses on organizing speech datasets for research related to voice authenticity and synthetic speech detection. As part of this work, datasets containing both real and generated (fake) voice samples were identified and collected. These datasets included speech samples of well-known personalities as well as synthesized speech generated using advanced speech models.

TRANSFORMER MODEL ANALYSIS MODULE:

This section examines transformer-based machine translation models, emphasizing large-scale designs with substantial parameter counts. Architectures approaching 1 billion parameters were analyzed to assess their multilingual proficiency, structural composition, and operational efficiency. Relevant research papers and technical specifications were reviewed to dissect encoder-decoder frameworks, attention mechanisms, and scalability features. Such insights clarified the reasons behind transformers' strong performance across India's varied linguistic spectrum.


facebook/nllb-200-distilled-600M
like 847
Follow AI at Meta 11.1k
Translation
Transformers
PyTorch
flores-200
196 languages
m2m_100
text2text-generation
nllb
License: cc-by-nc-4.0
Model card
Files and versions
xet
Community 46
NLLB-200
Downloads last month
265,669

4. CONCLUSION AND FUTURE SCOPE

During my two-week internship at Resilience Business Grid (RBG.AI), I explored two compelling AI domains: multilingual machine translation tailored for Indian languages and voice authenticity studies. My analysis of Hugging Face translation models revealed notable disparities in performance between high-resource languages (such as Hindi and Bengali) and low-resource ones (including many Dravidian tongues), underscoring that achieving fully inclusive AI for India's linguistic variety remains an ongoing endeavor.

In parallel, I contributed to assembling real and synthetic voice datasets for deepfake detection efforts. This experience highlighted the escalating importance of audio security amid increasingly sophisticated AI-generated speech.

In summary, the internship illuminated India's distinctive position in AI advancement—brimming with potential from its vast array of languages and dialects, yet facing substantial hurdles in developing equitable tools that serve beyond dominant languages.

Moving forward, enhancing support for low-resource Indian languages presents a prime opportunity, potentially through transfer learning from linguistically similar languages and data augmentation strategies to boost translation efficacy in data-poor settings.

Moreover, bolstering voice security measures grows ever more urgent. With synthetic audio becoming readily producible, prioritizing robust real-time deepfake detection and innovative protections like audio watermarking will prove vital to preserving confidence in digital communications.

5. LESSONS I HAVE LEARNED

My two-week internship at Resilience Business Grid (RBG.AI) offered intensive, hands-on learning that far exceeded my classroom studies. Here are the primary takeaways:

1. **Professional Research Workflow**: I contributed to an active AI research project with seasoned professionals, gaining insight into industry processes—from defining problems and reviewing literature to executing evaluations and delivering results. This taught me the value of meticulous documentation, accuracy, and aligning efforts with company objectives.
2. **Key AI Development Platforms**: I mastered Hugging Face for discovering, testing, and deploying cutting-edge transformer translation models, while leveraging GitHub for version control, experiment tracking, and building reproducible research repositories—essential tools for modern collaborative AI work.
3. **Power of Team Collaboration**: Working as part of a coordinated team underscored the necessity of task delegation (like model testing and data organization), knowledge sharing, and constant communication to unify diverse inputs into a unified outcome, proving that complex AI challenges thrive on collective expertise.
4. **Rigorous Model Assessment**: I advanced from basic model execution to in-depth comparative analysis using metrics like BLEU scores, learning to prioritize data-backed judgments over hype—evaluating fit for tasks like translating high-resource (e.g., Hindi) versus low-resource Indian languages.

5. **Data Preparation's Vital Role**: Handling multilingual speech and text datasets revealed how crucial high-quality curation and preprocessing are; subpar data undermines even the best models, forming the bedrock of successful AI outcomes.

6. INTERACTION WITH CO-WORKERS

I share a strong rapport with my supervisor and colleagues at Resilience Business Grid. The company's workplace culture is exceptionally supportive and inclusive, fostering an environment where I could freely engage with my boss and teammates whenever needed. They proved far more collaborative than I expected.

My mentors consistently offered help whenever I faced challenges, making me feel at ease when seeking guidance or sharing ideas. Once, I hesitated to voice a basic concept due to lacking a creative angle, but my supervisor encouraged me by saying every idea holds world-changing potential if refined properly. After I explained it, he provided a key insight I had missed, leading to further brainstorming that produced the strongest solution.

Mr. Parameshwaran B N, in particular, always guided me effectively on my tasks—clarifying what to do, how to approach it, and what to avoid—while checking for any concerns. The team's encouraging and approachable demeanor helped build my confidence and enabled meaningful contributions. I feel truly grateful for their warm interactions and constructive feedback at Resilience Business Grid.

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7. DIFFICULTIES ENCOUNTERED

One of the greatest aspects of the internship was the wealth of practical, real-world exposure it offered, helping bridge the transition from student life to professional maturity. I faced minimal issues at work thanks to my supportive and friendly colleagues and supervisor.

My primary difficulty involved adjusting to prolonged screen time—sitting at a computer for over eight hours daily was new to me, causing back pain, eye strain, and frequent headaches during the initial weeks. I also struggled with juggling multiple tasks within tight deadlines.

However, with guidance from my supervisor and teammates, I learned effective strategies to manage these situations and complete work more efficiently. Their patient mentoring enabled me to tackle unfamiliar challenges successfully without any major task-related hurdles, providing ample time and clear instructions to meet all requirements.

8. WEB REFERENCES

- *https://ai4bharat.iitm.ac.in/*
- https://huggingface.co/facebook/nllb-200-distilled-600M
- *https://commonvoice.mozilla.org/*
- *https://arxiv.org/abs/1810.04805*
- *https://aclanthology.org/P02-1040/*